# Detailed Preregistration for 'Quantifying Change: Challenges and Strategies in Effect Size Determination for Exercise Intervention Meta-Analyses'

Simon Nolte

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*This document includes the preregistration of the aforementioned project. It accompanies the 'Preregistration Summary' to provide more details of the planed work. This document follows the 'Inclusive Systematic Review Registration Form' (Akker et al., 2023) and will be published on the Open Science Framework.* *This preregistration is licensed under a CC-BY 4.0 license.*

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## Meta-data

### Target discipline

Sport Science; Sports Medicine; Meta-Research

### Title

'Quantifying Change: Challenges and Strategies in Effect Size Determination for Exercise Intervention Meta-Analyses'

### Authors / Contributors

*TO-DO*

### Tasks and Roles

*TO-DO*

Conceptualization; Data curation; Formal Analysis; Software; Investigation; Methodology; Visualization; Writing;

## Review methods

### Type of Review

Methodological review, with additional analysis of an example case (see Section ).

### Review Stages

Search, screening, data extraction

### Current review stage

Before search; the project has not started yet.

### Start date

01 Jan 2025

### End date

31 Mar 2025 (for data extraction)

## Background

Meta-analyses are frequently among the most cited research items in sports science and play a crucial role in guiding training practices and informing future research. However, methodological flaws in sports science meta-analyses are common (Hill, 2023; Kadlec et al., 2023). A frequently overlooked but potentially important aspect in many exercise intervention meta-analyses is the method used to calculate effect sizes and their confidence intervals for studies with with continuous outcomes and two groups (e.g., treatment and control) that include pre- and post-measurements.

For analyzing group comparisons with pre- and post-measurements, three methods are commonly used: ANCOVA, change score analysis, and post score analysis (Fu & Holmer, 2016). Each of these approaches varies in statistical power based on specific data characteristics and relies on different input variables. These input variables may be reported in the primary literature or need to be imputed for meta-analysis purposes. In the absence of universal guidelines, diverse approaches and imputation parameters may be employed in meta-analytical practice.

To date, no systematic review exists of the methods used to determine effect sizes in published exercise intervention meta-analyses. Since different analytical methods can lead to varying conclusions (Fu & Holmer, 2016), the use of a variety of methods may indicate a lack of robustness in many meta-research findings within exercise science. A systematic review, coupled with a review of available methodological literature, could provide valuable recommendations for sports scientists on calculating effect sizes in exercise intervention meta-analyses.

## Primary research question

Which methods do researchers use for individual study effect size calculation in two-group pre-post meta-analyses of exercise interventions?

## Secondary research questions

How do researchers report and justify their used method of effect size calculation? Can changing the effect size calculation method change an analysis' conclusion?

## Expectations / hypotheses

No formal hypotheses are tested. We anticipate that researchers use a variety of methods in the published literature, which are sometimes incompletely reported or inadequately justified. We expect that the choice of method can influence the results of a meta-analysis, with different methods potentially yielding varying overall effect estimates and significance levels.

### Dependent variable(s) / outcome(s) / main variable(s)

Effect size determination method used.

### Independent variable(s) / interventions(s) / treatment(s)

— No independent variables used —

### Additional variable(s) / covariate(s)

- Effect size determination method reported
- Justification for used method of effect size determination
- Criteria/assessment of baseline imbalances for studies included in the meta-analysis
- Effect size measure (mean difference, standardized mean difference)

### Software

For search and data analysis: R version 4.4.0 (or newer) (R Core Team, 2023), `meta` package for R, version 8.0 (or newer) (Balduzzi et al., 2019), `rentrez` package for R, version 1.2.3 (or newer) (Winter, 2017).

For screening and data extraction processing we use a self-developed web platform based on PHP and MySQL that coordinates the screening and extraction process.

### Funding

— No funding was received for this project —

### Conflicts of interest

— All researchers involved state that there are no conflicts of interest —

### Overlapping authorship

— No overlapping authorship is expected —

## Search Strategy

### Databases

PubMed

### Interfaces

PubMed via the `rentrez` package (Winter, 2017).

### Grey literature

— No grey literature will be searched —

### Inclusion and exclusion criteria

Inclusion:

- Only research published in one of the journals that published the 20 most cited meta-analyses in strength and conditioning research according to Kadlec et al. (2023).
- Only research published in 2024 (date of print publication). Note that this may be expanded based on the number of included articles (see Section ).
- Only research that contains the phrase "Meta-Analysis" in its title.

### Query strings

```
((("_JOURNALNAME1_"[Journal]) OR ("_JOURNALNAME2_)) AND (("2024/01/01"[Date - Publication] : "2024/12/31"[Date - Publication]))) AND (Meta-Analysis[Title])
```

Replace `_JOURNALNAMEX_` by the name of each journal. Adjust date range if necessary (see section Section )

### Search validation

We will determine the number of retrieved articles per journal and check the number against the number of articles matching the same criteria on the journal websites.

### Other search strategies

— No further search strategies are used —

### Procedures to contact authors

We will not contact any author during the review. We may contact authors during the case study (see Section ).

### Results of contacting authors

— Not applicable as we do not contact authors during the review —

### Search expiration and repetition

We will conduct one single literature search in January 2025, which will not be repeated.

### Search strategy justification

Our aim is to map current methodological practices in exercise science meta-analyses. We include only recent articles, as developments in software and methodological knowledge may have led to changes in the literature, and we seek to depict the current state of the field. We search for meta-analyses within a subset of sports science journals that we expect to represent high-quality, current research, as indicated by their publication of highly cited meta-analyses in strength and conditioning research (Kadlec et al., 2023). To focus on relevant studies, we limit our search to articles with the phrase "Meta-Analysis" in the title, assuming this terminology is standard for meta-analyses and helps filter out systematic reviews and original studies that do not contain a meta-analysis.

## Screening

### Screening stages

1. Title and abstract screening

2. Full-text retrieval and screening

   *If less than 100 results remain after step 2, we will extend the date range of the search by one previous year. This step may be repeated until a total of 100 results remain.*

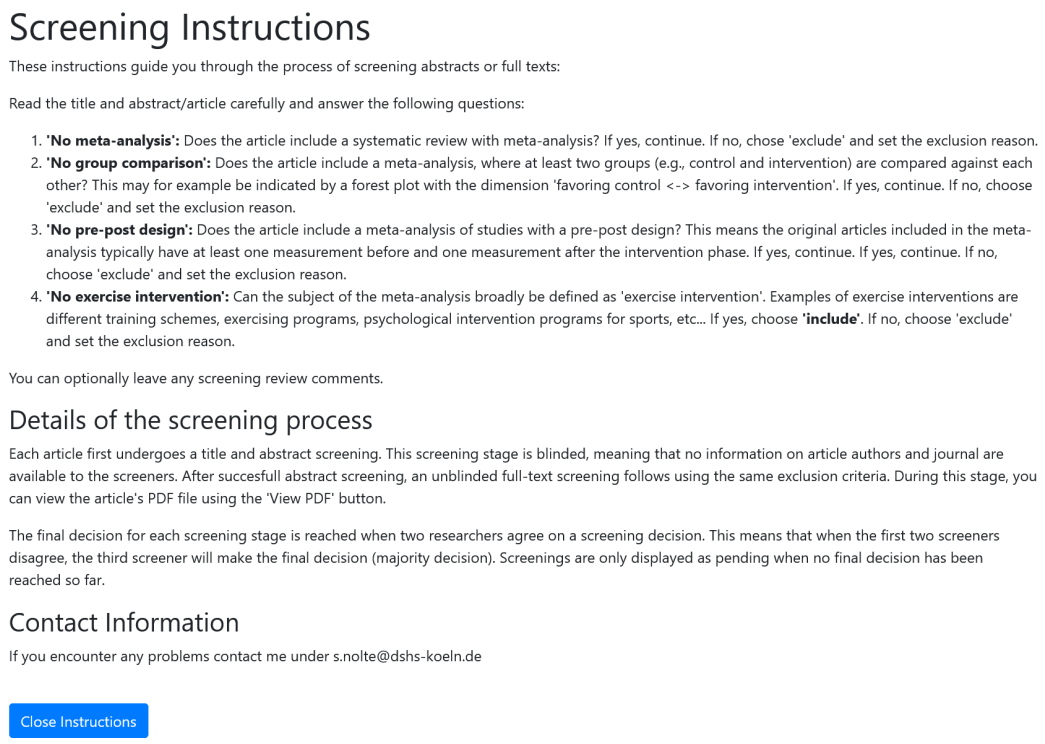
3. Data extraction

### Screened fields / blinding

During title and abstract screening, all additional information will be blinded to the person screening (screeners only see the article title and abstract, but not the journal, authors, or publication year). During full text screening, no blinding applies.

### Used exclusion criteria

See Figure 1.



## Screening Instructions

These instructions guide you through the process of screening abstracts or full texts:

Read the title and abstract/article carefully and answer the following questions:

1. **'No meta-analysis':** Does the article include a systematic review with meta-analysis? If yes, continue. If no, chose 'exclude' and set the exclusion reason.
2. **'No group comparison':** Does the article include a meta-analysis, where at least two groups (e.g., control and intervention) are compared against each other? This may for example be indicated by a forest plot with the dimension 'favoring control <-> favoring intervention'. If yes, continue. If no, choose 'exclude' and set the exclusion reason.
3. **'No pre-post design':** Does the article include a meta-analysis of studies with a pre-post design? This means the original articles included in the meta-analysis typically have at least one measurement before and one measurement after the intervention phase. If yes, continue. If yes, continue. If no, choose 'exclude' and set the exclusion reason.
4. **'No exercise intervention':** Can the subject of the meta-analysis broadly be defined as 'exercise intervention'. Examples of exercise interventions are different training schemes, exercising programs, psychological intervention programs for sports, etc... If yes, choose **'include'**. If no, choose 'exclude' and set the exclusion reason.

You can optionally leave any screening review comments.

### Details of the screening process

Each article first undergoes a title and abstract screening. This screening stage is blinded, meaning that no information on article authors and journal are available to the screeners. After succesfull abstract screening, an unblinded full-text screening follows using the same exclusion criteria. During this stage, you can view the article's PDF file using the 'View PDF' button.

The final decision for each screening stage is reached when two researchers agree on a screening decision. This means that when the first two screeners disagree, the third screener will make the final decision (majority decision). Screenings are only displayed as pending when no final decision has been reached so far.

### Contact Information

If you encounter any problems contact me under s.nolte@dshs-koeln.de

Close Instructions

Figure 1: Screening criteria and instructions in the review web platform

### Screener instructions

See Figure 1.

### Screening reliability

All screening will be employed by two independent screeners (researchers involved in the project can assign themselves to screening any article for which the screening result is pending). Screening reliability will be calculated based as percentage of consensus in initial decision (include/exclude).

### Screening reconciliation procedure

If the two screeners disagree in their screening decision (include/exclude) a third independent researcher will screen the article and the final result will be based on the majority vote.

### Sampling and sampling size

We aim for a sample size of ≥100 articles to allow for a sufficient subset of current meta-analysis research in exercise science. If after all screening steps less than 100 articles remain, we will extend the search period by one previous year, until the threshold of 100 articles after final exclusion is meet.

### Screening procedure justification

We only include articles that match the research design we are investigating (meta-analyses of group comparisons with pre-post measurements) and the general topic of the project (exercise interventions). The work of two to three instructed independent screeners helps to ensure a reliable and valid screening procedure.

### Data management and sharing

We will store the initial search results in a `csv` file. Results of screening and data extraction will be stored in the data base back-end of the web platform and can be exported. All data and analysis code will be made publicly available on GitHub.

### Miscellaneous screening details

When they later encounter errors in their screening, researchers can update their screening results. This will restart the screening process for the respective article.

## Extraction

### Entities to extract

See Figure 2.

### Extraction stages

Data extraction is conducted in three stages:

1. **Full-text data extraction:** One researcher extracts the reported characteristics of each article.
2. **Recalculation:** The same researcher attempts to recalculate an effect size for one study included in the meta-analysis. This study is selected as the first (top) study in the first forest plot that meets the screening criteria (i.e., a forest plot representing a pre-post group comparison design). If a different study is used for the reproducibility check, the reason is documented. If individual study effect sizes are not presented in the meta-analysis, recalculation is deemed infeasible.
3. **Validation:** An independent researcher reviews the extraction results. Using an annotated article full text, this researcher decides whether to validate the initial data extraction or to suggest changes. If changes are suggested, the article returns to Stage 1, where the original researcher re-evaluates their extraction results.

### Extractor instructions

See Figure 2.

### Extractor blinding

The researcher performing the validation is blinded to who performed the data extraction. Otherwise, no blinding applies.

# Extraction Instructions

These instructions guide you through the process of data screening and screening validation:

## Extraction Items: Full Text

Read the full text article carefully and answer the following questions:

| Item Name | Item Description | Possible Values |
|---|---|---|
| Reported Effect Size Calculation Method | What method does the meta-analysis *report* to define the study-level effect size measure. | 'ANCOVA', 'post-measures', 'change scores' or 'unreported'. |
| Reported ANCOVA Source | For ANCOVA-based effect sizes only: What method does the meta-analysis *report* to base its ANCOVA estimates on? | 'Individual Patient Data', 'Study-level ANCOVA estimates', or 'unreported'. |
| Reported Method to Calculate Change Score Standard Deviations | For change score-based effect sizes only: What method does the meta-analysis *report* to base the standard deviation of its study-level effects on? | Free text, e.g., 'Use reported change score SD where possible, use pre and post SD with imputed correlation instead' or 'unreported' |
| Reported Method to Impute the Pre-Post Correlation | For change score-based effect sizes only: What method does the meta-analysis *report* for the imputation of pre-post correlations for change score standard deviations? | Free text, e.g., 'use r = 0.5 for all studies' or 'unreported'. |
| Justification of the Effect Size Calculation (and Imputation) Method Used | What justification does the article provide for its choice of effect size measure, and calculation parameters? | Free text, e.g., 'reference to Cuipers et al. (2017)' or 'none'. |
| Effect Size Used | What effect size is the meta-analyses based on? | 'Mean Difference' or 'Standardized Mean Difference' |
| Baseline Imbalance Status | How did the meta-analysis assess the possibility of baseline imbalances in the included studies? | Free text, e.g., 'only randomized controlled trials included', 'randomization of individual studies rated as low risk of bias', or 'unclear' |
| Extraction Comments | Insert any further observations you made during the data extraction. | Free text |

## Extraction Items: Recalculation

Select one effect size from the first forest plot and try to recalculate it based on the original study data. Answer the following questions:

| Item Name | Item Description | Possible Values |
|---|---|---|
| Study Used for Recalculation | Which study was used for the recalculation attempt. | Free text, e.g., 'Extract et al. (2017); first study from forest plot (Fig. 2)'. |
| Effect Size Calculation Method | What method does the meta-analysis actually *use* to define the study-level effect size measure. | 'ANCOVA', 'post-measures', 'change scores', or 'unclear'. |
| ANCOVA Source | For ANCOVA-based effect sizes only: What method does the meta-analysis actually *use* to base its ANCOVA estimates on? | 'Individual Patient Data', 'Study-level ANCOVA estimates', or 'unclear'. |
| Method to Calculate Change Score Standard Deviations | For change score-based effect sizes only: What method does the meta-analysis actually *use* to base the standard deviation of its study-level effects on? *This might be difficult to answer based on a single original study.* | Free text, e.g., 'Use reported change score SD where possible, use pre and post SD with imputed correlation instead' or 'unclear' |
| Method to Impute the Pre-Post Correlation | For change score-based effect sizes only: What method does the meta-analysis actually *use* for the imputation of pre-post correlations for change score standard deviations? | Free text, e.g., 'r = 0.5' or 'unclear'. |
| Recalculation Comments | Insert any further observations you made during the effect size recalculation. | Free text |

## Details of the Data Extraction Process

Each included article takes part in the data extraction process. One researchers first answers the extraction questions based on the reporting in the article PDF. She will make annotations in the PDF to ease validation.

The same researcher then selects a random effect size from the first forest plot of the meta analysis and tries to recalculate the given study effect size based on the original research article. The results of this recalculation are documented in the recalculation section of the data extraction form. Again, remarks can be made in the annotated PDF. The researcher then submits the extraction report.

Another researcher validates the extraction report by reviewing it on the basis of the annotated PDF. She can decide to either validate or not validate the data extraction, and leave comments.

If the data extraction for an article has been validated, the extraction process for this article is completed. If an article gets the status 'not validated', the original author of the data extraction needs to re-check the extraction and make modifications if necessary.

The modified data extraction can be validated again by the author of the original validation.

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Close Instructions

Figure 2: Extraction criteria and instructions in the review web platform

### Extraction reliability

We perform a validation stage to check for potential issues in the data extraction process.

### Extraction reconciliation

If, during the validation phase, researchers find the data extraction results erroneous, they choose not to validate the extraction and instead leave comments for the original extractor. The extractor can then update the data extraction, which will subsequently be re-evaluated by the same researcher who performed the initial validation.

### Extraction procedure justification

For economic reasons, a single researcher will perform the initial data extraction. To minimize errors, we use an independent validation procedure. Additionally, we implement a recalculation stage to ensure that the reported analytical characteristics align with those actually applied in the included studies.

### Data management and sharing

The data extraction results will be stored in the data base back-end of the web platform and can be exported. The exported data will be made available on GitHub.

## Synthesis and quality assessment

As this is a methodological (scoping) review, we will not perform any systematic quality assessment of the included research. Instead we will count the frequency of specific methods used for effect size determination. Additionally, we will calculate the percentage of studies with missing, incomplete, incorrect, or correct reporting of their methods. The justifications provided for the chosen methods will be qualitatively assessed. Finally, we will exploratively analyze, whether the choice of methods varies depending on how baseline imbalances are addressed.

## Example case

We will use an example meta-analysis that matches the review inclusion criteria to investigate whether different effect size determination methods can change the results of a meta-analysis.

### Study selection

For the case study, we use the article by Oliveira et al. (2024), as it served as the motivating example for the project lead (SN) to initiate this research project. While reviewing the meta-analysis, he identified a triple-counting of studies in the forest plot of the article's main finding (Fig. 4 in Oliveira et al. (2024)). Additionally, attempts to reproduce the meta-analysis revealed discrepancies, as the standard deviations of change scores reported in the meta-analysis could not be directly reproduced. Through communication with the lead author, he was able to reconstruct the (previously unreported) method used by the authors. However, this raised further questions about whether altering the methods for effect size calculation might influence the conclusions of the meta-analysis.

### Data Extraction

We will extract all reported summary measures from the original studies included in Fig. 4 of Oliveira et al. (2024). For missing summary measures (e.g., pre-post correlations or change score standard deviations) and individual data points, we will attempt to obtain the necessary information by extracting data from figures and/or contacting the authors of the respective articles.

To derive a reasonable set of imputation parameters for pre-post correlations, we will extract these measures from a set of studies investigating exercise interventions targeting the same outcome parameter as Oliveira et al. (2024) (maximum oxygen uptake). This set of studies is drawn from the analysis conducted by Renwick et al. (2024). Specifically, we will extract or recalculate all pre-post correlations from the studies included in Renwick et al. (2024) to determine informed imputation parameters (e.g., the median pre-post correlation) and to establish a full distribution of correlations.

### Reanalysis

We will recalculate the meta-analysis presented in Fig. 4 of Oliveira et al. (2024) after correcting for the multiple counting of effect sizes. This will involve combining groups when multiple control groups exist and selecting the most frequently studied exercise type when multiple exercise conditions are reported for a single study. Except for the method of effect size calculation, we will follow the analytical decisions made by Oliveira et al. (2024), such as using the standardized mean difference and a random-effects meta-analysis.

Our recalculation will apply different approaches to effect size calculation, including ANCOVA, post scores, and change scores with different imputation parameters. The specific set of methods employed will depend on data availability and the findings of the systematic review, which will indicate prominent methods in current research.

For each effect size calculation method, we will calculate the overall mean effect size and its 95%-confidence interval. These results will be compared to evaluate the original research hypothesis by determining whether the confidence intervals overlap with an effect size of zero.

## References


Akker, O. R. van den, Peters, G.-J. Y., Bakker, C. J., Carlsson, R., Coles, N. A., Corker, K. S., Feldman, G., Moreau, D., Nordström, T., Pickering, J. S., Riegelman, A., Topor, M. K., Veggel, N. van, Yeung, S. K., Call, M., Mellor, D. T., & Pfeiffer, N. (2023). Increasing the transparency of systematic reviews: Presenting a generalized registration form. *Systematic Reviews*, *12*(1), 170. https://doi.org/10.1186/s13643-023-02281-7

Balduzzi, S., Rücker, G., & Schwarzer, G. (2019). How to perform a meta-analysis with R: a practical tutorial. *BMJ Ment Health*, *22*(4), 153–160. https://doi.org/10.1136/ebmental-2019-300117

Fu, R., & Holmer, H. K. (2016). Change score or follow-up score? Choice of mean difference estimates could impact meta-analysis conclusions. *Journal of Clinical Epidemiology*, *76*, 108–117. https://doi.org/10.1016/j.jclinepi.2016.01.034

Hill, A. P. (2023). Comparisons and Conversions: A Methodological Note and Caution for Meta-Analysis in Sport and Exercise Psychology. *Journal of Sport and Exercise Psychology*, *45*(5), 293–296. https://doi.org/10.1123/jsep.2023-0076

Kadlec, D., Sainani, K. L., & Nimphius, S. (2023). With Great Power Comes Great Responsibility: Common Errors in Meta-Analyses and Meta-Regressions in Strength & Conditioning Research. *Sports Medicine*, *53*(2), 313–325. https://doi.org/10.1007/s40279-022-01766-0

Oliveira, P. S., Boppre, G., & Fonseca, H. (2024). Comparison of Polarized Versus Other Types of Endurance Training Intensity Distribution on Athletes' Endurance Performance: A Systematic Review with Meta-analysis. *Sports Medicine*, *54*(8), 2071–2095. https://doi.org/10.1007/s40279-024-02034-z

R Core Team. (2023). *R: A language and environment for statistical computing*. https://www.R-project.org/

Renwick, J. R. M., Preobrazenski, N., Wu, Z., Khansari, A., LeBouedec, M. A., Nuttall, J. M. G., Bancroft, K. R., Simpson-Stairs, N., Swinton, P. A., & Gurd, B. J. (2024). Standard Deviation of Individual Response for VO2max Following Exercise Interventions: A Systematic Review and Meta-analysis. *Sports Medicine*. https://doi.org/10.1007/s40279-024-02089-y

Winter, D. J. (2017). rentrez: An R package for the NCBI eUtils API. *The R Journal*, *9*(2), 520–526. https://doi.org/10.32614/RJ-2017-058